Idan Steinberg, Ph.D

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Dynamic, highly innovative engineer with expertise in medical devices, medical imaging, and signal processing.
Over 25 years of hands-on experience in research positions at internationally renowned institutions.
Proven ability to design and execute studies to elucidate the efficacy of complex products. Excel in identifying product opportunities for emerging technologies. Personable and engaging with excellent communication skills.

EXPERIENCE

Senior Director of Research and Development / Vice President for Research and Development

Endra Life Sciences Inc. **December 2024 - Present, Ann Arbor, MI**

- Strategic Technical Leadership - TRL 1: proposed a generation system design that reduced console size by 5x, probe weight by 5x, addressed a critical performance metric, and reduced cost by 33%.
- Regulatory & Clinical Support - Design, approval, and successful execution of multisite feasibility studies and R&R studies.
- Operational and execution responsibility for all company products - Spearheaded cross-functional initiatives to advance regulatory submission strategies, aligning research milestones with commercial objectives, and enabling efficient technology transfer to clinical and engineering teams.
- Product Development & Engineering - Led the technical team in a successful compilation of TRL 0 to TRL 5 in one year and on a lean budget, renovating the company's product and addressing critical performance issues.

Applied Science Team Leader

Endra Life Sciences Inc. **September 2021 - November 2024, Ann Arbor, MI**

- System-level ownership over the research and development of a Thermo-acoustic device for non-invasive liver fat assessment - Set target performance, minimum performance, and risk management at the system level.
- Team & Resource Management - Unified cross-disciplinary teams (Ann Arbor main office, London, ON offsite team, and Starfish medical engineering contractor) and allocated resources efficiently to support project timelines, ensuring alignment with organizational objectives and regulatory requirements.
- Key contributor to over 20 IP disclosures and technical resources to support commercial, engineering, and regulatory teams.
- Directed on-site experimental validation activities across 24 months, managing biweekly device performance tests and ensuring compliance with ISO 13485 standards.
- Initiated and maintained an academic collaboration program and brought on board 3 key laboratories from top universities across the globe (UIUC, UCL, and Technion).

Principal Researcher

Vortex Medical Imaging **November 2020 – September 2021, Sunnyvale, CA**

- Guiding the development of the flagship product - a 2D low-frequency transducer array for full waveform volumetric abdominal imaging.
- Infrastructure and acoustic lab bring-up in less than 1 quarter: phantom fabrication, acoustic measurements and simulations, coordination with multiple subcontractors, and software platform development.
- In charge of determining specifications, developing new QA test procedures for the product, and employee training.
- Working both onsite (Sunnyvale) and at a collaborator ultrasound laboratory at Stanford to efficiently utilize resources.

Team Leader

Stanford Univ. School of Medicine, Radiology Dept. **January 2018 - May 2021, Palo Alto, CA**

- Guided cross-functional teams of 10-20 multidisciplinary researchers and engineers in optics and acoustics groups across 40 projects over 5 years.
- Provided both guidance and prioritization as an experienced professional, and assisted the PI in decision-making and early identification of successes and failures.
- Personally led 10 and actively participated in 40 concurrent projects spanning Raman preclinical imaging, targeted photoacoustic contrast agents, nanoscale acoustic contrast agents, RF-acoustic imaging, cancer imaging, and clinical trials.

Postdoctoral Fellow

Stanford Univ. School of Medicine, Radiology Dept. **June 2016 - May 2021, Palo Alto, CA**

- Developed a highly complex photoacoustic and ultrasound prostate imaging system from concept to feasibility clinical trial over the course of 5 years and a budget of $1M.
- Engineered custom signal processing algorithms and software workflows using MATLAB, Python, and CUDA for real-time photoacoustic data analysis to accelerate imaging rate by 10-fold.
- Conceptualized, created the initial design, identified and managed subcontractors in prototyping new hardware to ensure quality and timeliness of delivery of probe and cart housing, multi-port optical waveguide, and ultrasound array.
- Designing and executing preclinical, 2 ex vivo studies, and a feasibility clinical trial for photoacoustic imaging applications for early cancer detection in collaboration with multidisciplinary academic and clinical teams.
- Project evaluated and selected as one of 5 awardees for funding by Philips Healthcare for high translational and commercial potential.
- Promoted to Team Leader after exceeding expectations through the development of creative solutions.

Additional work experience or letters of reference will be available on request

EDUCATION

Accelerated Management Development Certificate Program

Univ. of Michigan, Ross School of Business • Ann Arbor, MI • 2023

- Analytical, value-driven, and people-driven thinking skill development, emphasizing strategic decision-making and digital innovation.
- Identifying key value drivers and opportunities for value creation, developing a strategic go-to-market plan, and understanding market dynamics to identify sources of competitive advantage.
- Negotiation and team building skills.

Ph.D., Biomedical Engineering

Tel Aviv Univ., EE & BME dept. • Tel Aviv, Israel • 2016

- Focus on acoustic instrumentation development, Biophotonics, and fiber optics.
- Leveraged ultrasound to assess bone strength and photoacoustics to detect early molecular signs of osteoporosis.
- Named one of nine Sir Charles Clore Scholars.

M.Sc., Biomedical Engineering & Management Track

Minor in Business Administration • Tel Aviv Univ., BME dept. • Tel Aviv, Israel • 2010

- Focus on statistical optics and laser-tissue interactions (Monte-Carlo simulations, diffusion approximation).
- Developed a device to generate acoustic waves in tumors labeled with magnetic nanoparticles.
- Selected for direct-track to M.Sc. program as top 5% of B.Sc. students - full scholarships.

B.Sc., Biomedical Engineering

Tel Aviv Univ., BME dept. • Tel Aviv, Israel • 2008

INVOLVEMENT

- International Society for Optics and Photonics at Tel Aviv Univ - Founder and first President.

- President's Peres program for nurturing Israel's future scientists and inventors at Tel Aviv Univ.

AWARDS & HONORS

- **Sir Peter Michael Foundation Fellow** - Named one of several national researchers selected as a Sir Peter Michael Foundation Fellow for excellence in clinical translation of prostate cancer photoacoustic imaging at Stanford University.

- **Philips Healthcare Fellows -** at Stanford University, Department of Radiology, for developing a dual Ultrasound/Photoacoustic system

- **SPARK Program Fellow -** Chosen as a SPARK Program Fellow as one of 14 Stanford teams for clinical translation of photoacoustic imaging of bacterial infections.

- **Sir Charles Clore Scholars** - Highly competitive program supporting the top 10 applicants nationwide.

PUBLICATIONS

Photoacoustics 14, 77-98

Photoacoustic clinical imaging

Superiorized Photo-Acoustic Non-Negative Reconstruction (SPANNER) for Clinical Photoacoustic Imaging

IEEE Transactions on Medical Imaging • 2021

First-in-human study of bone pathologies using a low-cost and compact dual-wavelength photoacoustic system

IEEE Journal of Selected Topics in Quantum Electronics • 2018

A new method for tumor detection using induced acoustic waves from tagged magnetic nanoparticles

Nanomedicine: Nanotechnology, Biology, and Medicine • 2012

SKILLS

Expertise: Data Analysis & Visualization, Biomedical Sensors, Biomedical Signal Processing, Human Physiology, Software Development, Multidisciplinary Collaboration

Programming: C++, Matlab, CUDA, Python, LabVIEW, Java, Delphi

CAD and simulation software: Rhino 3D, ZeeMax, Comsol, Photoshop, 3D Studio Max, Adobe Premiere, Monte Carlo simulations, k-Wave, and Field II acoustic simulations

Hardware experience: Free space and fiber-optics, RF, and acoustic lab instrumentation. Control, automation, and remote data acquisition (DAQ). Specialization in building prototype acoustic and optical devices, testing, and calibration

Personal: Frontal and active teaching. Mentoring and team leadership. My superpower is my ability to learn, adapt, and excel